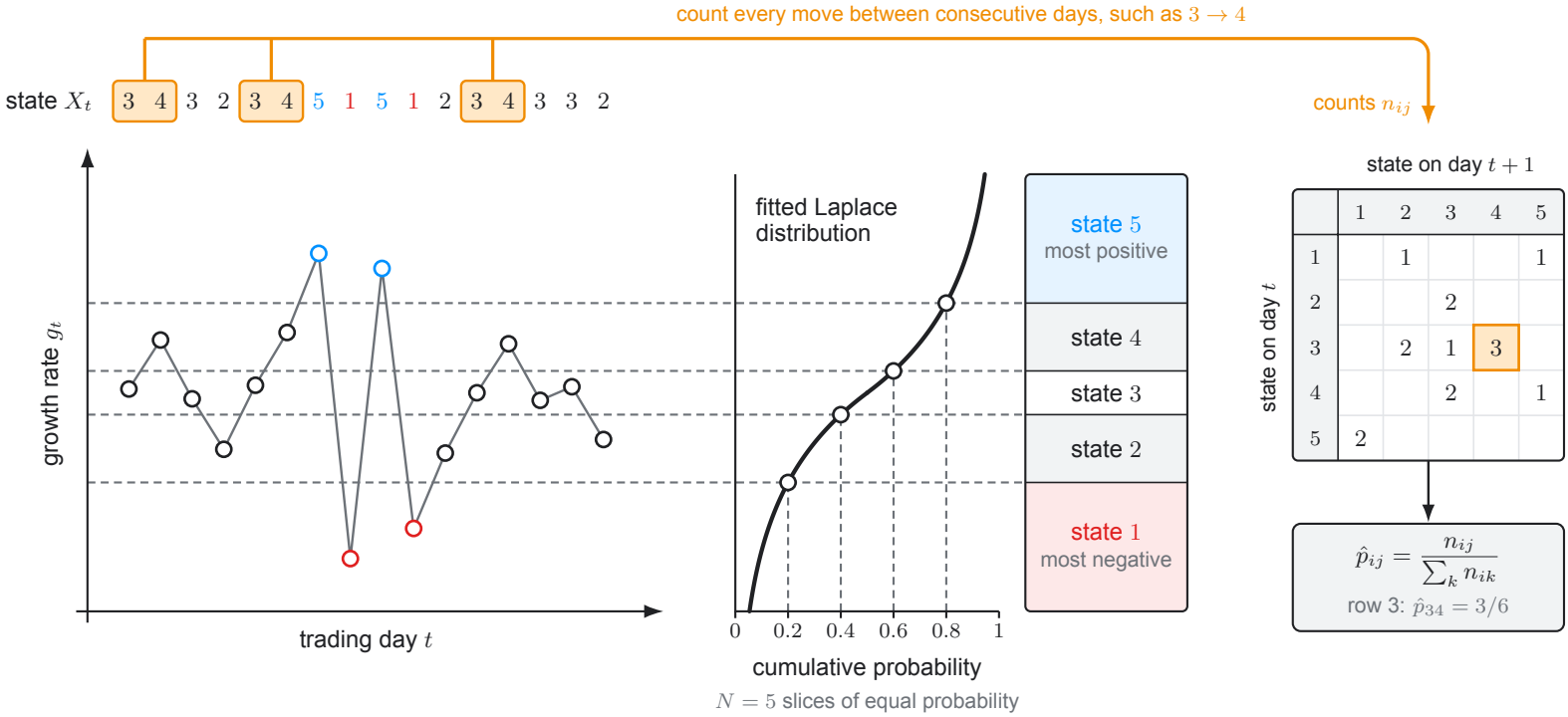
count every move between consecutive days, such as $3 \to 4$
state X_t
3 4 3 2 3 4 5 1 5 1 2 3 4 3 3 2
counts n_{ij}
growth rate g_t
trading day t
fitted Laplace distribution
state 5
most positive
state 4
state 3
state 2
state 1
most negative
0 0.2 0.4 0.6 0.8 1
cumulative probability
$N = 5$ slices of equal probability
state on day $t+1$
state on day t
1 2 3 4 5
1 1 1
2 2
3 2 1 3
4 2 1
5 2
$\hat{p}_{ij} = \frac{n_{ij}}{\sum_k n_{ik}}$
row 3: $\hat{p}_{34} = 3/6$